

ATTENUATOR CODE LOADED WITH ZO:

```
import numpy as np
import matplotlib.pyplot as plt

# =====
# 1) Theoretical V2/V1 transfer
# =====

R1 = 47.0
R2 = 47.0
R3 = 760.0
R4 = 3.3
RL = 50.0

num = RL * (R1*R2 + R1*R4 + R2*R4 + R3*R4)
den = (R1*R2*R3 + R1*R2*RL +
       R1*R3*R4 + R1*R3*RL + R1*R4*RL +
       R2*R3*R4 + R2*R4*RL +
       R3*R4*RL)

H = num / den
mag_db_teo = 20 * np.log10(abs(H))

print("H = V2/V1 =", H)
print("Gain in dB (theoretical) =", mag_db_teo)

# ===== FREQUENCY RANGE = MEASURES =====
f_min = 50.0      # Hz
f_max = 5000.0    # Hz

# Theoretical curve ONLY in the range of measurements
f = np.logspace(np.log10(f_min), np.log10(f_max), 1000)
mag_db_teo_vec = mag_db_teo * np.ones_like(f)

# =====
# 2) Measured data
# =====

f_with = np.array([50, 100, 250, 500, 750, 900, 1000, 2500, 3000,
                   5000], dtype=float)
mag_db_med = np.array([-23.3, -22.66, -23.63, -23.3, -23.99,
                       -23.19, -23.19, -23.27, -23.85, -23.52])

# =====
# 3) Comparative graph
# =====

plt.figure()

plt.semilogx(f, mag_db_teo_vec, label='Simulation / Theoretical',
             linewidth=2)
```

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plt.semilogx(f_with, mag_db_med, 'the-', label='Laboratory
measurement', markersize=6)

plt.xlabel('Frequency [Hz]')
plt.ylabel('Magnitude [dB]')
plt.title('V2/V1 Comparison: Simulation vs Measurement')
plt.grid(True, which='both', ls='--')

# >>> HERE WE SET THE X-AXIS RANGE <<<
plt.xlim(f_min, f_max)

plt.ylim(mag_db_teo - 3, mag_db_teo + 3)
plt.legend()
plt.tight_layout()
plt.show()

```

FILTER CODE LOADED WITH ZO:

```

import numpy as e.g
import matplotlib.pyplot as plt

# =====
# 1) SIMULATION:  $H(s) = V2 / V1$  (theoretical)

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# =====

R1 =47.0
R2 =47.0
RL = 51.0
L1 =7.58e-6
L2 =7.45e-6
C1 =3.3e-9
C2 = 3.3e-9

# Coefficients of H(s)
a3 =(C1*C2*L2*R1*R2*RL + C1*L1*L2*RL)
a2 =(C1*L2*R1*RL + C1*L2*R2*RL)
a1 =(C1*R1*R2*RL + L2*RL)
a0 =0.0

b4 =(C1*C2*L1*L2*R1*R2 + C1*C2*L1*L2*R1*RL)
b3 =(C1*C2*L2*R1*R2*RL + C1*L1*L2*R1 + C1*L1*L2*R2 + C1*L1*L2*RL)
b2 =(C1*L1*R1*R2 + C1*L1*R1*RL + C1*L2*R1*RL + C1*L2*R2*RL
      + C2*L2*R1*R2 + C2*L2*R1*RL)
b1 =(C1*R1*R2*RL + L2*R1 + L2*R2 + L2*RL)
b0 =(R1*R2 + R1*RL)

# Frequency sweep: 100 kHz to 5 MHz
f_teo = np.logspace(5, np.log10(5e6), 1000) # 1e5 ... 5e6 Hz
in =2 * np.pi * f_theo
s = 1j * In

number = a3*s**3 + a2*s**2+ a1*s + a0
den = b4*s**4+ b3*s**3 + b2*s**2+ b1*s + b0
H = num / den

att_db_teo =20 * np.log10(e.g.abs(H))

# =====
# 2) LABORATORY MEASUREMENTS
# =====

f_khz = np.array([
    100, 200, 350, 400, 500, 650, 700, 800, 950,
    1000, 1100, 1250, 1300, 1400, 1550, 1600, 1700,
    1850, 1900, 2000, 2150, 2200, 2300, 2450,
    2500, 2800, 3000, 3400, 3800, 4000, 4400, 4800, 5000
], dtype=float)

f_med = f_khz * 1e3

att_db_with = np.array([
    -19.89, -14.02, -8.57, -7.83, -5.87, -3.67, -2.88, -1.97,
    -1.09,
    -0.71, -0.55, -1.17, -1.05, -1.36, -1.96, -2.55, -3.00,
    -3.68, -3.86, -4.31, -5.23, -5.74, -5.51, -6.37,
    -6.48, -7.15, -7.96, -9.15, -9.54, -9.89, -10.75, -11.79,
    -11.39
])

```

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])

# =====
#3) COMPARATIVE GRAPH
# =====

plt.figure()

plt.semilogx(f_teo, att_db_teo, label='Simulation V2/V1',
linewidth=2)
plt.semilogx(f_with, att_db_med, 'the-', label='Laboratory
measurement', markersize=6)

plt.xlabel('Frequency [Hz]')
plt.ylabel('Attenuation [dB]')
plt.title('Band-pass filter: simulation vs measurement')
plt.grid(True, which='both', ls='--')

# Frequency range: 100 kHz to 5 MHz
plt.xlim(1e5, 5e6)

plt.legend()
plt.tight_layout()
plt.show()

```

FILTER CODE LOADED WITH ZO 10 TIMES GREATER:

```

import numpy as e.g
import matplotlib.pyplot as plt

# =====
# 1) LABORATORY MEASUREMENTS
# =====

f_khz_med = np.array([10, 100, 500, 800, 1000, 1500,
                      1800, 2000, 3000, 4000, 5000], dtype=float)
f_with = f_khz_with *1e3 # Hz

mag_db_med = np.array([
    -31.97, -15.35, -0.63, 0.74, -0.30,
    1.23, 1.40, 0.95, -2.96, -5.69, -8.20

```

```

])

# =====
# 2) SIMULATION:  $H(s) = V2 / V1$  ( $R_L = 510 \Omega$ )
# =====

R1 = 47.0
R2 = 47.0
RL = 510.0      # Zo 10 times larger
L1 = 7.58e-6
L2 = 7.45e-6
C1 = 3.3e-9
C2 = 3.3e-9

# Coefficients of H(s)
a3 = (C1*C2*L2*R1*R2*RL + C1*L1*L2*RL)
a2 = (C1*L2*R1*RL + C1*L2*R2*RL)
a1 = (C1*R1*R2*RL + L2*RL)
a0 = 0.0

b4 = (C1*C2*L1*L2*R1*R2 + C1*C2*L1*L2*R1*RL)
b3 = (C1*C2*L2*R1*R2*RL + C1*L1*L2*R1 + C1*L1*L2*R2 + C1*L1*L2*RL)
b2 = (C1*L1*R1*R2 + C1*L1*R1*RL + C1*L2*R1*RL + C1*L2*R2*RL
      + C2*L2*R1*R2 + C2*L2*R1*RL)
b1 = (C1*R1*R2*RL + L2*R1 + L2*R2 + L2*RL)
b0 = (R1*R2 + R1*RL)

# Frequency sweep: 10 kHz to 5 MHz
f_teo = np.logspace(np.log10(1e4), np.log10(5e6), 1000)
in_ = 2 * np.pi * f_theo
s = 1j * in_

number = a3*s**3 + a2*s**2 + a1*s + a0
den = b4*s**4 + b3*s**3 + b2*s**2 + b1*s + b0
H = num / den

mag_db_teo = 20 * np.log10(e.g.abs(H))

# =====
#3) COMPARATIVE GRAPH
# =====

plt.figure()

# Simulation
plt.semilogx(f_teo, mag_db_teo, label='Simulation V2/V1 (RL = 510
 $\Omega$ )', linewidth=2)

# Measurements
plt.semilogx(f_with, mag_db_med, 'the-', label='Laboratory
measurement', markersize=6)

plt.xlabel('Frequency [Hz]')
plt.ylabel('Magnitude / Attenuation [dB]')

```

```

plt.title('Band-pass filter loaded with RL = 510  $\Omega$ : simulation vs
measurement')
plt.grid(True, which='both', ls='--')

plt.xlim(1e4, 5e6)    # 10 kHz a 5 MHz

plt.legend()
plt.tight_layout()
plt.show()

```

FILTER CODE LOADED WITH ZO 10 TIMES SMALLER:

```

import numpy as e.g
import matplotlib.pyplot as plt

# =====
# 1) LABORATORY MEASUREMENTS (RL = 5  $\Omega$ )
# =====

f_khz_med = np.array([10, 100, 500, 800, 1000, 1500,
                      1800, 2000, 3000, 4000, 5000], dtype=float)
f_with = f_khz_with *1e3    # Hz

att_db_with = np.array([
    -45.66, -33.62, -20.93, -15.09, -5.90,
    -16.42, -18.81, -19.96, -24.08, -26.02, -27.73
])

# =====
# 2) SIMULATION: H(s) = V2 / V1 (RL = 5  $\Omega$ )
# =====

R1 =47.0
R2 =47.0

```

```

RL = 5.0          # Zo 10 times smaller
L1 = 7.58e-6
L2 = 7.45e-6
C1 = 3.3e-9
C2 = 3.3e-9

# Coefficients of H(s)
a3 = (C1*C2*L2*R1*R2*RL + C1*L1*L2*RL)
a2 = (C1*L2*R1*RL + C1*L2*R2*RL)
a1 = (C1*R1*R2*RL + L2*RL)
a0 = 0.0

b4 = (C1*C2*L1*L2*R1*R2 + C1*C2*L1*L2*R1*RL)
b3 = (C1*C2*L2*R1*R2*RL + C1*L1*L2*R1 + C1*L1*L2*R2 + C1*L1*L2*RL)
b2 = (C1*L1*R1*R2 + C1*L1*R1*RL + C1*L2*R1*RL + C1*L2*R2*RL
      + C2*L2*R1*R2 + C2*L2*R1*RL)
b1 = (C1*R1*R2*RL + L2*R1 + L2*R2 + L2*RL)
b0 = (R1*R2 + R1*RL)

# Frequency sweep: 10 kHz to 5 MHz
f_teo = np.logspace(np.log10(1e4), np.log10(5e6), 1000) # Hz
in = 2 * np.pi * f_theo
s = 1j * in

number = a3*s**3 + a2*s**2 + a1*s + a0
den = b4*s**4 + b3*s**3 + b2*s**2 + b1*s + b0
H = num / den

att_db_teo = 20 * np.log10(np.abs(H))

# =====
#3) COMPARATIVE GRAPH
# =====

plt.figure()

# Simulation
plt.semilogx(f_teo, att_db_teo, label='Simulation V2/V1 (RL = 5 Ω)',
             linewidth=2)

# Measurements
plt.semilogx(f_with, att_db_med, 'the-', label='Laboratory
measurement', markersize=6)

plt.xlabel('Frequency [Hz]')
plt.ylabel('Attenuation [dB]')
plt.title('Bandpass filter loaded with RL = 5 Ω: simulation vs
measurement')
plt.grid(True, which='both', ls='--')

plt.xlim(1e4, 5e6) # 10 kHz a 5 MHz

plt.legend()
plt.tight_layout()

```

```
plt.show()
```

FILTER CODE LOADED WITH ZO AND WITH THE ATTENUATOR BEFORE:

```
import numpy as e.g
import matplotlib.pyplot as plt

# =====
# 1) SIMULATION: complete circuit V2/V1
# (attenuator + band-pass filter)
# =====

# Attenuator (V3/V1)
R1a =47.0
R2a =47.0
R3a =760.0
R4a =3.3
RLa = 50.0    # "ideal" 50 Ω load seen by the attenuator

num_att = RLa * (R1a*R2a + R1a*R4a + R2a*R4a + R3a*R4a)
that_that = (R1a*R2a*R3a + R1a*R2a*RLa +
R1a*R3a*R4a + R1a*R3a*RLa + R1a*R4a*RLa +
R2a*R3a*R4a + R2a*R4a*RLa +
R3a*R4a*RLa)

H_att = num_att / den_att    # complex constant (real)

# Filtro LC (V2/V3)
R1 =47.0
R2 =47.0
RL = 51.0
L1 =7.58e-6
L2 =7.45e-6
C1 =3.3e-9
C2 = 3.3e-9
```



```

a3 =(C1*C2*L2*R1*R2*RL + C1*L1*L2*RL)
a2 =(C1*L2*R1*RL + C1*L2*R2*RL)
a1 =(C1*R1*R2*RL + L2*RL)
a0 =0.0

b4 =(C1*C2*L1*L2*R1*R2 + C1*C2*L1*L2*R1*RL)
b3 =(C1*C2*L2*R1*R2*RL + C1*L1*L2*R1 + C1*L1*L2*R2 + C1*L1*L2*RL)
b2 =(C1*L1*R1*R2 + C1*L1*R1*RL + C1*L2*R1*RL + C1*L2*R2*RL
      + C2*L2*R1*R2 + C2*L2*R1*RL)
b1 =(C1*R1*R2*RL + L2*R1 + L2*R2 + L2*RL)
b0 =(R1*R2 + R1*RL)

# Frequency sweep 10 kHz - 5 MHz
f_teo = np.logspace(np.log10(1e4), np.log10(5e6), 1000)
in =2 * np.pi * f_theo
s = 1j * In

num_f = a3*s**3 + a2*s**2+ a1*s + a0
den_f = b4*s**4+ b3*s**3 + b2*s**2+ b1*s + b0
H_f = num_f / den_f

# Circuito completo: V2/V1 = H_att * H_f
H_tot = H_att * H_f
mag_tot_db =20 * np.log10(e.g.abs(H_tot))

# =====
# 2) LABORATORY MEASUREMENTS
# =====

f_khz_med = np.array([10, 100, 500, 800, 1000,
                      1500, 1800, 2000, 3000, 4000, 5000],
dtype=float)
f_with = f_khz_with *1e3 # Hz

att_db_with = np.array([
    -44.39, -41.88, -29.25, -25.26, -24.36,
    -25.41, -27.17, -28.62, -32.22, -31.54, -38.64
])

# =====
#3) COMPARATIVE GRAPH
# =====

plt.figure()

# Simulation
plt.semilogx(f_teo,mag_tot_db,
              label='Simulation V2/V1 (attenuator + filter)',
linewidth=2)

# Measurements
plt.semilogx(f_with,att_db_med,
              'the-', label='Laboratory measurement', markersize=6)

```

```
plt.xlabel('Frequency [Hz]')
plt.ylabel('Attenuation [dB]')
plt.title('Band-pass filter + attenuator: simulation vs
measurement')
plt.grid(True, which='both', ls='--')

plt.xlim(1e4, 5e6) # 10 kHz a 5 MHz

plt.legend()
plt.tight_layout()
plt.show()
```

FILTER CODE LOADED WITH ZO 10 TIMES LARGER AND WITH THE ATTENUATOR BEFORE:

```
import numpy as e.g
import matplotlib.pyplot as plt

# =====
# 1) SIMULATION: complete circuit V2/V1
#   (H(s) already obtained for RL = 510 Ω)
# =====

# Coefficients of H(s) = V2/V1
a3 = 9.37278723856467e-13
a2 = 5.92508255328e-06
a1 = 37.7907041184

b4 = 1.31182072376078e-18
b3 = 1.64572775124139e-11
b2 = 1.58251494561547e-04
b1 = 665.9548264048
b0 = 2.1331486928e9

# Frequency sweep: 10 kHz to 5 MHz
f_teo = np.logspace(np.log10(1e4), np.log10(5e6), 2000)
in_ = 2 * np.pi * f_theo
s = 1j * In

number = a3*s**3 + a2*s**2 + a1*s
den = b4*s**4 + b3*s**3 + b2*s**2 + b1*s + b0
H = num / den

mag_db_teo = 20 * np.log10(e.g.abs(H))

# =====
# 2) LABORATORY MEASUREMENTS
# =====

# Frequency in kHz (according to the table)
f_khz_med = np.array([10, 100, 500, 800, 1000,
                      1500, 1800, 2000, 3000, 4000, 5000],
dtype=float)
f_with = f_khz_with * 1e3 # Hz

# Attenuation in dB
att_db_with = np.array([
```

```

-44.68, -36.55, -24.53, -20.77, -20.16,
-20.51, -22.21, -23.51, -28.21, -31.62, -31.68
])

# =====
#3) COMPARATIVE GRAPH
# =====

plt.figure()

# Simulation
plt.semilogx(f_teo, mag_db_teo,
              label='Simulation V2/V1 (RL = 510  $\Omega$ )', linewidth=2)

# Measurements
plt.semilogx(f_with, att_db_med,
              'the-', label='Laboratory measurement', markersize=6)

plt.xlabel('Frequency [Hz]')
plt.ylabel('Attenuation [dB]')
plt.title('Band-pass filter + attenuator (RL = 510  $\Omega$ ): simulation
vs measurement')
plt.grid(True, which='both', ls='--')

plt.xlim(1e4, 5e6) # 10 kHz a 5 MHz

plt.legend()
plt.tight_layout()
plt.show()

```

FILTER CODE LOADED WITH ZO 10 TIMES SMALLER AND WITH THE ATTENUATOR BEFORE:

```
import numpy as e.g
import matplotlib.pyplot as plt

# =====
# 1) SIMULATION: complete circuit V2/V1 (RL = 5 Ω)
# =====

# Coefficients of H(s) = V2/V1 for the complete circuit with RL = 5
Ω
a3 = 9.18900709663200e-15
a2 = 5.80890446400000e-08
a1 = 3.70497099200000e-01
a0 = 0.0

b4 = 1.22468002936374e-19
b3 = 1.56118754890600e-12
b2 = 1.47728247434592e-05
b1 = 6.27283155736000e+01
b0 = 1.99144940800000e+08

# Frequency sweep: 10 kHz to 5 MHz
f_teo = np.logspace(np.log10(1e4), np.log10(5e6), 2000)
in_ = 2 * np.pi * f_theo
s = 1j * In

number = a3*s**3 + a2*s**2 + a1*s + a0
den = b4*s**4 + b3*s**3 + b2*s**2 + b1*s + b0
H = num / den

mag_db_teo = 20 * np.log10(e.g.abs(H))

# =====
# 2) LABORATORY MEASUREMENTS (RL = 5 Ω)
# =====

f_khz_med = np.array([
    10, 100, 500, 800,
    1000, 1500, 1800, 2000,
    3000, 4000, 5000
], dtype=float)

f_with = f_khz_with * 1e3 # Hz

att_db_with = np.array([
    -40.81, -39.50, -40.41, -34.09,
    -33.22, -36.98, -40.17, -41.31,
    -43.49, -39.34, -37.08
])
```

```

# =====
#3) COMPARATIVE GRAPH
# =====

plt.figure()

# Simulation
plt.semilogx(f_teo, mag_db_teo,
              label='Simulation V2/V1 (RL = 5  $\Omega$ )', linewidth=2)

# Measurements
plt.semilogx(f_with, att_db_med,
              'the-', label='Laboratory measurement', markersize=6)

plt.xlabel('Frequency [Hz]')
plt.ylabel('Attenuation [dB]')
plt.title('Band-pass filter + attenuator - RL = 5  $\Omega$ : simulation vs measurement')
plt.grid(True, which='both', ls='--')

plt.xlim(1e4, 5e6) # 10 kHz a 5 MHz

plt.legend()
plt.tight_layout()
plt.show()

```